

```
# Heart Disease Dataset
import pandas as pd
```

```
df=pd.read_csv('/content/heart_disease_uci (1).csv')
df.head()
```

	id	age	sex	dataset	cp	trestbps	chol	fbs	restecg	thalch	exang	oldpeak	slope	ca	thal	r
0	1	63	Male	Cleveland	typical angina	145.0	233.0	True	lv hypertrophy	150.0	False	2.3	downsloping	0.0	fixed defect	
1	2	67	Male	Cleveland	asymptomatic	160.0	286.0	False	lv hypertrophy	108.0	True	1.5	flat	3.0	normal	
2	3	67	Male	Cleveland	asymptomatic	120.0	229.0	False	lv hypertrophy	129.0	True	2.6	flat	2.0	reversible defect	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.tail()
```

	id	age	sex	dataset	cp	trestbps	chol	fbs	restecg	thalch	exang	oldpeak	slope	ca	thal	num
915	916	54	Female	VA Long Beach	asymptomatic	127.0	333.0	True	st-t abnormality	154.0	False	0.0	NaN	NaN	NaN	1
916	917	62	Male	VA Long Beach	typical angina	NaN	139.0	False	st-t abnormality	NaN	NaN	NaN	NaN	NaN	NaN	0
917	918	55	Male	VA Long Beach	asymptomatic	122.0	223.0	True	st-t abnormality	100.0	False	0.0	NaN	NaN	fixed defect	2

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```
df.shape
```

(920, 16)

```
df.describe()
```

	id	age	trestbps	chol	thalch	oldpeak	ca	num	
count	920.000000	920.000000	861.000000	890.000000	865.000000	858.000000	309.000000	920.000000	
mean	460.500000	53.510870	132.132404	199.130337	137.545665	0.878788	0.676375	0.995652	
std	265.725422	9.424685	19.066070	110.780810	25.926276	1.091226	0.935653	1.142693	
min	1.000000	28.000000	0.000000	0.000000	60.000000	-2.600000	0.000000	0.000000	
25%	230.750000	47.000000	120.000000	175.000000	120.000000	0.000000	0.000000	0.000000	
50%	460.500000	54.000000	130.000000	223.000000	140.000000	0.500000	0.000000	1.000000	
75%	690.250000	60.000000	140.000000	268.000000	157.000000	1.500000	1.000000	2.000000	
max	920.000000	77.000000	200.000000	603.000000	202.000000	6.200000	3.000000	4.000000	

```
df.columns
```

Index(['id', 'age', 'sex', 'dataset', 'cp', 'trestbps', 'chol', 'fbs',
 'restecg', 'thalch'],
 dtype='object')

```
df.tail()
```

```
chol
915 333.0
916 139.0
917 223.0
918 385.0
919 254.0

dtype: float64
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 920 entries, 0 to 919
Data columns (total 16 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   id           920 non-null    int64
1   age          920 non-null    int64
2   sex          920 non-null    object
3   dataset      920 non-null    object
4   cp           920 non-null    object
5   trestbps     861 non-null    float64
6   chol         890 non-null    float64
7   fbs         830 non-null    object
8   restecg      918 non-null    object
9   thalch       865 non-null    float64
10  exang        865 non-null    object
11  oldpeak      858 non-null    float64
12  slope        611 non-null    object
13  ca           309 non-null    float64
14  thal         434 non-null    object
15  num          920 non-null    int64
dtypes: float64(5), int64(3), object(8)
memory usage: 115.1+ KB
```

```
df.isnull().sum()
```

```
0
id      0
age      0
sex      0
dataset  0
cp       0
trestbps 59
chol     30
fbs      90
restecg  2
thalch   55

dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 920 entries, 0 to 919
Data columns (total 16 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   id           920 non-null    int64
1   age          920 non-null    int64
2   sex          920 non-null    object
3   dataset      920 non-null    object
```

```
4 cp 920 non-null object
5 trestbps 920 non-null float64
6 chol 920 non-null float64
7 fbs 920 non-null bool
8 restecg 920 non-null object
9 thalch 920 non-null float64
10 exang 865 non-null object
11 oldpeak 858 non-null float64
12 slope 611 non-null object
13 ca 309 non-null float64
14 thal 434 non-null object
15 num 920 non-null int64
dtypes: bool(1), float64(5), int64(3), object(7)
memory usage: 108.8+ KB
```

```
df['trestbps'] = df['trestbps'].fillna(df['trestbps'].mean())
```

```
df['trestbps'].isnull().sum()
```

```
np.int64(0)
```

```
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	30
fbs	90
restecg	2
thalch	55
exang	55
oldpeak	62
slope	309
ca	611
thal	486
num	0

dtype: int64

```
df['chol'] = df['chol'].fillna(df['trestbps'].mean())
```

```
df['chol'].isnull().sum()
```

```
np.int64(0)
```

```
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	90
restecg	2
thalch	55
exang	55
oldpeak	62
slope	309
ca	611
thal	486
num	0

dtype: int64

```
df['fbs'] = df['fbs'].fillna(df['fbs'].mode()[0])
```

```
df['fbs'].isnull().sum()
```

```
np.int64(0)
```

```
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	2
thalch	55
exang	55
oldpeak	62
slope	309
ca	611
thal	486
num	0

dtype: int64

```
df['restecg'] = df['restecg'].fillna(df['restecg'].mode()[0])
```

```
df['restecg'].isnull().sum()
```

```
np.int64(0)
```

```
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	55
exang	55
oldpeak	62
slope	309
ca	611
thal	486
num	0

```
dtype: int64
```

```
df['thalch'] = df['thalch'].fillna(df['thalch'].mean())
```

```
df['thalch'].isnull().sum()
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	0
exang	55
oldpeak	62
slope	309
ca	611
thal	486
num	0

```
dtype: int64
```

```
df['exang'] = df['exang'].fillna(df['exang'].mode()[0])
df['exang'].isnull().sum()
df.isnull().sum()
```

```
/tmp/ipykernel_2352/1319527350.py:1: FutureWarning: Downcasting object dtype arrays on .fillna, .ffill, .bfill is deprecated and
df['exang'] = df['exang'].fillna(df['exang'].mode()[0])
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	0
exang	0
oldpeak	62
slope	309
ca	611
thal	486
num	0

dtype: int64

```
df['slope'] = df['slope'].fillna(df['slope'].mode()[0])
df['slope'].isnull().sum()
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	0
exang	0
oldpeak	62
slope	0
ca	611
thal	486
num	0

dtype: int64

```
df['thal'] = df['thal'].fillna(df['thal'].mode()[0])
df['thal'].isnull().sum()
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	0
exang	0
oldpeak	62
slope	0
ca	611
thal	0
num	0

dtype: int64

```
df['oldpeak'] = df['oldpeak'].fillna(df['oldpeak'].mean())

df['oldpeak'].isnull().sum()
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	0
exang	0
oldpeak	0
slope	0
ca	611
thal	0
num	0

dtype: int64

```
df['ca'] = df['ca'].fillna(df['ca'].mode()[0])
df['ca'].isnull().sum()
df.isnull().sum()
```

	0
id	0
age	0
sex	0
dataset	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
num	0

dtype: int64

```
'''sex
dataset
cp
fbs
restecg
exang
slope
thal'''
```

```
df['sex'].unique()
df['sex'].value_counts()
```

	count
sex	
Male	726
Female	194

dtype: int64

```
df['dataset'].unique()
df['dataset'].value_counts()
```

	count
dataset	
Cleveland	304
Hungary	293
VA Long Beach	200
Switzerland	123

dtype: int64

```
df['cp'].unique()
df['cp'].value_counts()
```

count	
cp	
asymptomatic	496
non-anginal	204
atypical angina	174
typical angina	46

dtype: int64

```
df['restecg'].unique()
df['restecg'].value_counts()
```

count	
restecg	
normal	553
lv hypertrophy	188
st-t abnormality	179

dtype: int64

```
df['fbs'].unique()
df['fbs'].value_counts()
```

count	
fbs	
False	782
True	138

dtype: int64

```
df['exang'].unique()
df['exang'].value_counts()
```

count	
exang	
False	583
True	337

dtype: int64

```
df['slope'].unique()
df['slope'].value_counts()
```

count	
slope	
flat	654
upsloping	203
downsloping	63

dtype: int64

```
df['thal'].unique()
df['thal'].value_counts()
```

count	
thal	
normal	682
reversable defect	192
fixed defect	46

dtype: int64

```
# Features (All columns except 'num')
X = df.drop('num', axis=1)
y = df['num']
```

```
X = df.drop('num', axis=1)
```

```
X.head()
```

	id	age	sex	dataset	cp	trestbps	chol	fbs	restecg	thalch	exang	oldpeak	slope	ca	thal
0	1	63	Male	Cleveland	typical angina	145.0	233.0	True	lv hypertrophy	150.0	False	2.3	downsloping	0.0	fixed defect
1	2	67	Male	Cleveland	asymptomatic	160.0	286.0	False	lv hypertrophy	108.0	True	1.5	flat	3.0	normal
2	3	67	Male	Cleveland	asymptomatic	120.0	229.0	False	lv hypertrophy	129.0	True	2.6	flat	2.0	reversable defect

Next steps: [Generate code with X](#) [New interactive sheet](#)

```
y.head()
```

num	
0	0
1	2
2	1
3	0
4	0

dtype: int64

```
print(X.shape)
print(y.shape)
```

```
(920, 15)
(920,)
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

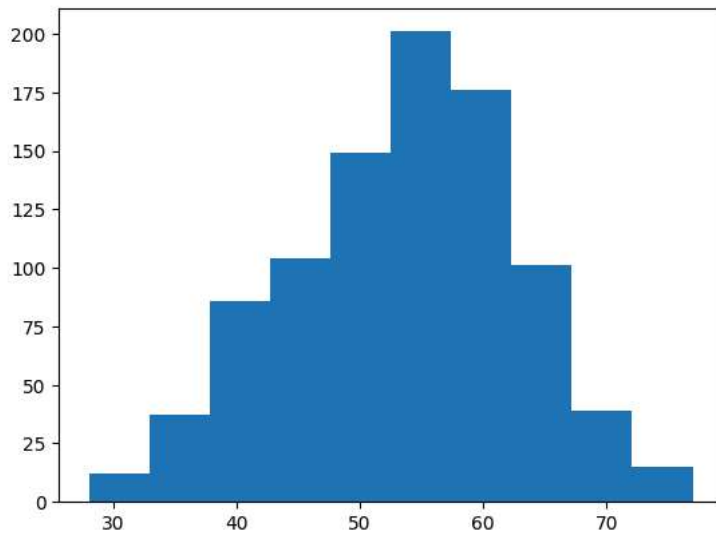
```
(736, 15)
(184, 15)
```

```
(736,)  
(184,)
```

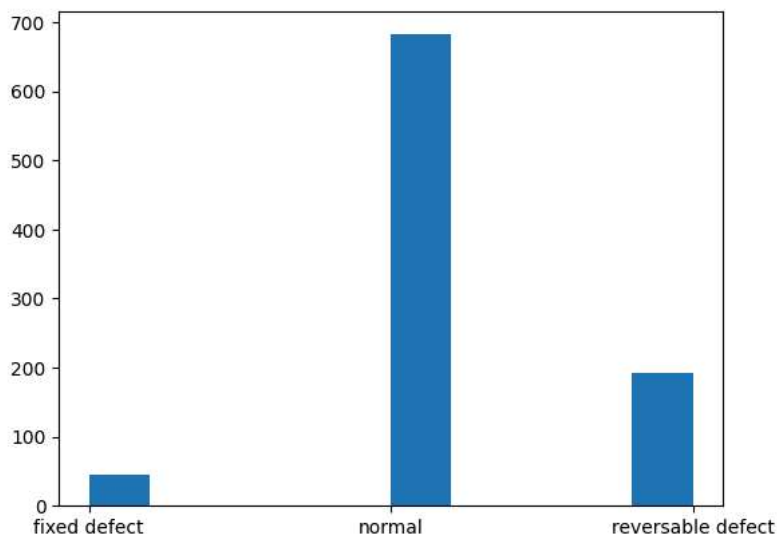
```
print(y.value_counts())
```

```
num  
0    411  
1    265  
2    109  
3    107  
4     28  
Name: count, dtype: int64
```

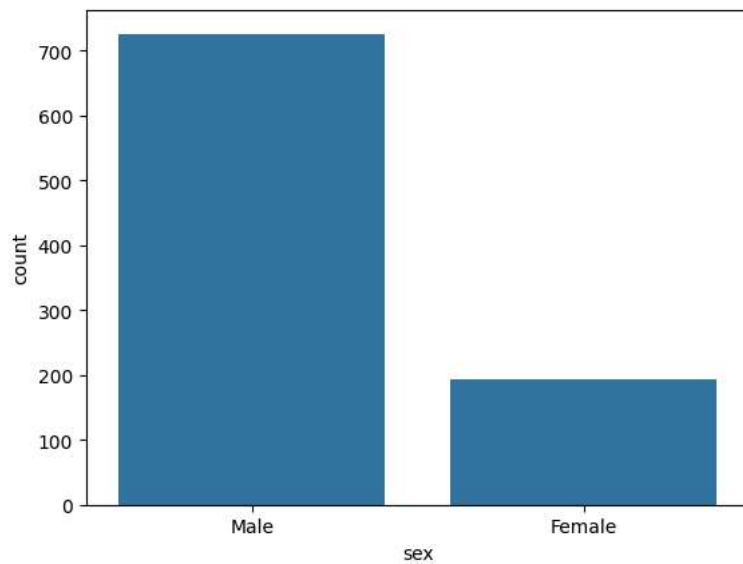
```
import matplotlib.pyplot as plt  
plt.hist(df['age'])  
plt.show()
```



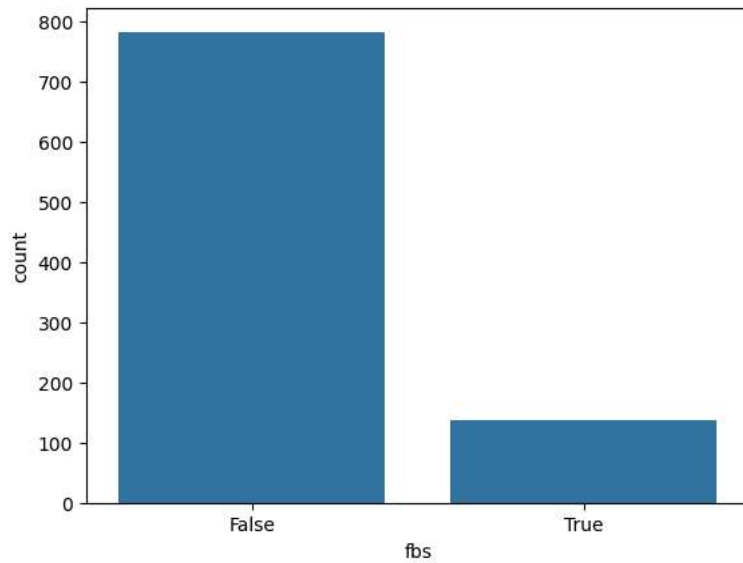
```
import matplotlib.pyplot as plt  
plt.hist(df['thal'])  
plt.show()
```



```
import seaborn as sns  
sns.countplot(x='sex', data=df)  
plt.show()
```



```
import seaborn as sns
sns.countplot(x='fbs',data=df)
plt.show()
```



```
sns.boxplot(x=df['chol'])
plt.show()
```

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.heatmap(df.corr(numeric_only=True))
plt.show()
```

